Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 60%. We keep your highest score.

Next item →

1. A developer is building an MCP-based AI application that runs inside an IDE. The application needs to translate user requests into a standardized format, manage session lifecycle events such as timeouts and reconnections, and route all interactions with external tools through a single interface.

Which MCP component is primarily responsible for these tasks?

1 / 1 point

- ○ The transport layer, because it handles authentication, message framing, and secure communication channels.
- ○ The MCP host, because it manages user input, applies interface controls, and presents AI responses.
- ○ The MCP server, because it executes tools, connects to external systems, and supplies contextual information to the AI.
- ◉ The MCP client, because it translates user requests into MCP-compliant JSON-RPC messages and manages communication and session handling with the server.

Nice work

The MCP client runs within the host, converts user requests into structured MCP messages, manages errors and sessions, and acts as the unified gateway to MCP servers.

2. What problem does the Model Context Protocol (MCP) primarily solve when building AI-powered applications?

1 / 1 point

- ◉ It replaces custom integrations and one-off adapters with a single standard interface for tools and data.
- ○ It allows AI models to generate code without any external context.
- ○ It ensures AI agents run only on local machines rather than the cloud.
- ○ It removes the need for AI models to use APIs entirely.

Nice work

MCP eliminates the need for bespoke glue code by acting like a universal connector for AI agents and external systems.

3. Which statement best describes the relationship between an MCP client and the tools exposed by an MCP server?

1 / 1 point

- ○ The MCP server only sends results, while the client is responsible for defining each tool.
- ○ The client contains all the tools locally and runs them without contacting the server.
- ◉ The client connects to the MCP server, discovers available tools, and calls them by passing the required parameters.
- ○ The tools run inside the client, while the MCP server only manages authentication.

Nice work

This reflects the MCP workflow. The client connects via a transport, lists available tools, and invokes them by name with parameters.

4. What is the primary benefit of using the MultiServerMCPClient when building an MCP application?

1 / 1 point

- ○ It replaces the need for an MCP client or agent.
- ○ It limits the number of connected MCP servers to prevent overload.
- ◉ It can connect to, and update, multiple MCP servers at the same time while handling low-level transport details.
- ○ It allows developers to avoid configuring transports entirely.

Nice work

MultiServerMCPClient supports simultaneous multi-server connections and manages the transport layer on behalf of the application.

5. What is the purpose of the InMemorySaver object in the MCP-powered agent system?

1 / 1 point

- ○ It stores MCP server configuration details for HTTP and STDIO transports.
- ○ It enables asynchronous execution of MCP server communication.
- ◉ It provides conversation memory so the agent can retain context across multiple exchanges.
- ○ It discovers tools available across connected MCP servers.

Nice work

InMemorySaver enables the agent to remember previous messages during the conversation.